

Practical 10 - Estimating causal effects using observational data 1

Survival Analysis

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Datasets required: rotterdam_p10

R packages required: survival, tableone, pec, adjustedCurves, riskRegression, mexhaz

```
library(survival)
library(tableone)
library(pec)
library(adjustedCurves)
library(riskRegression)
library(mexhaz)
```

Introduction

In this practical you will perform analyses to obtain the results shown in the lecture, using the Rotterdam breast cancer data. The Rotterdam data comprises data on 2982 individuals who underwent surgery for primary breast cancer and whose data were recorded in the Rotterdam Tumour Bank. Participants were followed-up for up to 20 years for the outcomes of disease recurrence and death from any cause. In this practical the focus is going to be on the composite outcome of disease recurrence or death without recurrence, and on the first 5 years of follow-up. The data have been modified from the original to apply administrative censoring at 5 years.

The variables we will use are summarised in the table below.

Variable name	Description
pid	Patient identifier
year_cat	Year of surgery. 0: before 1990, 1: 1990 or later

Variable name	Description
age	Age at surgery
meno	Menopausal status. 0=premenopausal, 1=postmenopausal
size	Tumour size: categorical with levels ≤ 20 mm, 20 – 50mm, > 50 mm
grade	Differentiation grade: categorical with levels 2 and 3
enodes	Number of positive lymph nodes, transformed as $\text{enodes} = \exp(-0.12 \cdot \text{nodes})$
pgr	Progesterone receptors (fmol/l)
er	Estrogen receptors (fmol/l)
chemo	Chemotherapy: 0=no, 1=yes
hormon	Hormonal treatment: 0=no, 1=yes
time	Years to the first of recurrence or death without recurrence
outcome	Indicator of composite outcome: 0=censored, 1=event
outcome_cr	Competing events outcome: 0=censored, 1=recurrence, 2=death without recurrence

Aims

The aims of this session are:

- Estimate conditional and marginal survival probabilities at a specific time under different treatment options.
- Estimate causal effects of a treatment in terms of marginal risk differences, risk ratios and restricted mean survival times.
- Estimate marginal survival curves under different treatment options.

Question 1

- Load the data.
- How many events are there and how many used hormone therapy?
- Compare the characteristics of users and non-users of hormone therapy. The `tableone` package is useful for this:

```
table1<-CreateTableOne(
  vars=c("year_cat","age","meno","size","grade","enodes",
        "pgr","er","hormon","chemo"),
  strata="hormon",
  data=roth,
  factorVars=c("year_cat","meno","size","grade","hormon","chemo"),
```

```

includeNA = FALSE,
test = F,
addOverall = T
)

print(table1,showAllLevels = T)

```

Question 2

- Fit an unadjusted Cox model including only the hormone therapy variable. Interpret the hazard ratio estimate.
- Fit a Cox model with hormone therapy as your main exposure, and conditioning on all other explanatory variables included in the data. The variables to include are: `age`, `meno`, `size`, `grade`, `enodes`, `pgr`, `er`, `year_cat`, `chemo`. We will assume that these are all potential confounders. Interpret the hazard ratio estimate for the hormone therapy variable and compare it with the unadjusted estimate.

Question 3

In this question we will use the adjusted model fitted in Question 2 to obtain an estimate of the risk of the composite event up to 5 years for each individual in the data under two treatment strategies:

- Start hormone therapy following surgery (and continue as recommended by the doctor) (`hormon=1`),
- Do not start hormone therapy following surgery (`hormon=0`).

Use the code below to do this for (i), and modify it for (ii).

What is the estimated risk of recurrence or death up to 5 years for person 1 in the data if they were to have followed treatment strategies (i) and (ii)?

```

#create dataset in which hormon is set to 1 for everyone
data.hormon1<-roth
data.hormon1$hormon<-1

#predicted survival probability at time 5 for each individual using cox.adj
#Using predictSurvProb from pec package
risk.hormon1.time5<-1-predictSurvProb(cox.adj,newdata=data.hormon1,times=5)

```

Question 4

In this question we will use the results from the previous question to obtain estimates of the marginal (adjusted) risks under the two treatment strategies.

- Obtain the marginal risk estimates $\Pr(T \leq 5|do(X = 1))$ and $\Pr(T \leq 5|do(X = 0))$, where X denotes hormone therapy strategy and T denotes the time of the composite event of interest. This can be done by taking the average of the conditional risk estimates obtained in question 3.
- Obtain estimates of the risk difference $\Pr(T \leq 5|do(X = 1)) - \Pr(T \leq 5|do(X = 0))$ and the risk ratio $\Pr(T \leq 5|do(X = 1)) / \Pr(T \leq 5|do(X = 0))$.

Question 5

- In this question we will use a ready-made function `adjustedsurv` from the `adjustedCurves` package to estimate the marginal (adjusted) risks differences and risk ratios, alongside confidence intervals. Take a look at the code below to do this.
- The estimates can be obtained using `adjsurv$ate_object`. Find the marginal risks and the marginal risk difference and risk ratio, and their 95% confidence intervals

```
#It turns out that `hormon` needs to be a factor variable for this
rottp$hormonfac<-as.factor(rottp$hormonfac)

#Refit Cox model with hormon as factor
#Also need to use x=TRUE here
cox.adj<-coxph(Surv(time,outcome)~hormonfac+age+meno+size+
               as.factor(grade)+enodes+pgr+er+year_cat+chemo,
               data=rottp,x=TRUE)

adjsurv <- adjustedsurv(data=rottp,
                       variable="hormonfac",
                       ev_time="time",
                       event="outcome",
                       method="direct",
                       outcome_model=cox.adj,
                       conf_int=TRUE,
                       bootstrap=F,times=5)
```

Question 6

So far we have focused on estimating the effect of hormone therapy on the risk up to time 5. We now extend this to obtain estimates of the marginal adjusted survival curves under the two treatment strategies.

- Use the code below to do this using `adjustedsurv`. The solution file shows how to do this without using `adjustedsurv`.
- What is the interpretation of the resulting curves?

```
adjsurv <- adjustedsurv(data=rott,
                        variable="hormonfac",
                        ev_time="time",
                        event="outcome",
                        method="direct",
                        outcome_model=cox.adjfac,
                        conf_int=TRUE,
                        bootstrap=F)

plot(adjsurv, conf_int=T, custom_colors=c("black", "grey"),
     xlab="Time (years)",
     legend.title="Hormone therapy", ylim=c(0,1))
```

Extra questions/examples

Question 7

In the above analyses we obtained marginal risks (and risk differences and risk ratios) based on standardisation of conditional estimated from the Cox model `cox.adj` (or equivalently `cox.adjfac`). This made the proportional hazards assumption for all variables included in the model.

In this question we relax the proportional hazards assumption for `hormon` by stratifying on this variable, i.e. by allowing a separate baseline hazard in the `hormon=0` and `hormon=1` groups. After fitting this modified model, the subsequent standardisation steps are the same as above.

See the code below for how to do this, and compare the results with those from Question 5.

```
#Refit Cox model stratified by hormone status
cox.adjstrat<-coxph(Surv(time,outcome)~strata(hormonfac)+age+meno+size+
                    as.factor(grade)+enodes+pgr+er+year_cat+chemo,
                    data=rott,x=TRUE)
```

```

adjsurv <- adjustedsurv(data=roth,
                        variable="hormonfac",
                        ev_time="time",
                        event="outcome",
                        method="direct",
                        outcome_model=cox.adjstrat,
                        conf_int=TRUE,
                        bootstrap=F, times=5)

#risks at time 5
adjsurv$ate_object$meanRisk

#risk diff at time 5
adjsurv$ate_object$diffRisk

#risk ratio at time 5
adjsurv$ate_object$ratioRisk

```

Question 8

Another estimand of interest could be the restricted mean survival times (RMST) under the two treatment strategies, or differences in the RMST. Based on the estimates of the marginal risks (or survival probabilities) obtained above, here we show how to estimate the RMST up to time 5 under the two treatment strategies, and their difference.

The code provided illustrates how to estimate the marginal RMST and RMST differences using `adjustedsurv`.

Interpret the RMST estimates and their difference.

```

#---
#Obtain RMST under each treatment strategy using adjustedCurves.
#Bootstrapping is required to get the confidence intervals for the RMST.
#Note you should use at least 500 bootstrap samples,
#but this is quite slow so we have just used 50 below for illustration.

adjsurv <- adjustedsurv(data=roth,
                        variable="hormonfac",
                        ev_time="time",
                        event="outcome",
                        method="direct",
                        outcome_model=cox.adjfac,

```

```

        conf_int=TRUE,
        bootstrap=T,n_boot=50)

#RMST under the two treatment strategies
adjrmst <- adjusted_rmst(adjsurv,
                        from=0, to=max(rott$time[rott$outcome==1]),
                        conf_int=TRUE,conf_level=0.95)

adjrmst

#RMST difference
adjrmst.diff <- adjusted_rmst(adjsurv,
                             from=0, to=max(rott$time[rott$outcome==1]),
                             conf_int=TRUE,conf_level=0.95,
                             contrast = "diff")

adjrmst.diff

```

Question 9

To estimate marginal risks using standardisation based on fully parametric models, such as a Weibull, you can use the `mexhaz` package, as shown below.

```

#fit the weibull model
mex.weibull<-mexhaz(formula=Surv(time,outcome)~hormon+age+meno+size+
                    as.factor(grade)+enodes+pgr+er+year_cat,
                    data=rott, expected = NULL, base = "weibull")

#obtain marginal survival probabilities at time 5, and their difference
adjsurv.weibull <- adjsurv(mex.weibull,time.pts=5,
                          data=data.hormon1,data.0=data.hormon0)

adjsurv.weibull

#adj.surv is Pr(T>5|do(X=1)), (adj.surv.inf, adj.surv.sup) is the 95% CI
#adj.surv.0 is Pr(T>5|do(X=0)), (adj.surv.0.inf, adj.surv.0.sup) is the 95% CI
#diff.adj.surv is Pr(T>5|do(X=1))-Pr(T>5|do(X=0)) (1 minus the risk diff),
# and (diff.adj.surv.inf,diff.adj.surv.sup) is the 95% CI

```